

Network Traffic Investigation

Professional Portfolio Report

Project Overview

A hands-on network investigation of a Windows endpoint using Wireshark and a real-world malware-traffic training PCAP. The project demonstrates an evidence-driven workflow from endpoint identification and protocol analysis through suspicious HTTP investigation, artifact extraction, and initial static analysis.

Scenario

The PCAP represented traffic from a Windows host in an Active Directory environment. The investigation was performed without executing the captured payload on the analysis workstation. Network evidence was used to attribute the endpoint and identify communications requiring further investigation.

Objectives

- Identify the investigated Windows endpoint and associated account.
- Analyze relevant DNS, TLS, HTTP, Kerberos, SMB, and SAMR traffic.
- Identify suspicious external communications and extract network artifacts.
- Perform an initial static assessment of the extracted artifact.

Tools & Technologies

Wireshark · 7-Zip · Detect It Easy (DIE) · FLOSS · Ghidra · Windows virtual machine

Endpoint Attribution

Evidence	Result	Method
IP Address	10.9.11.135	Wireshark Endpoints / traffic correlation
MAC Address	08:d4:0c:7a:29:1e	NBNS / Ethernet evidence
Hostname	DESKTOP-6T17ZFM	Kerberos traffic
User Account	gmcdownell	Kerberos CNameString
Account Full Name	Gabriel McDowell	SAMR QueryUserInfo response

Network Investigation

Endpoint and conversation statistics were used to identify the primary investigative host. DNS traffic showed queries from the endpoint to the internal DNS server, including a resolution for **quadcinema.com** to **104.196.13.170**. TLS analysis also identified an SNI value of **nodejs.org** in a separate encrypted connection. These observations were treated as individual evidence points rather than automatically classified as malicious.

HTTP Investigation

Clear-text HTTP traffic provided a stronger investigation lead. The endpoint communicated with **86.106.87.134** using the host **know.mom-nower.com**. The traffic included requests carrying **application/octet-stream** data. A large POST transfer was examined using TCP-stream analysis, and HTTP objects were exported from the PCAP for offline investigation.

Extracted Artifact

The largest relevant HTTP object was extracted as **hfwniayzki**, approximately 4.45 MB in size. The artifact was handled offline and transferred into an isolated Windows analysis environment for static examination.

Initial Static Analysis

The artifact was examined with Detect It Easy, FLOSS, and Ghidra. The sample did not present a confirmed standard Windows PE executable structure during the initial analysis. String and disassembly inspection also did not provide sufficient evidence to establish a definitive malware classification.

Key Indicators of Interest

Type	Indicator
Internal Endpoint	10.9.11.135
External IP	86.106.87.134
HTTP Host	know.mom-nower.com
Artifact	hfwniayzki

Assessment

The investigation identified suspicious network communications and a suspicious HTTP-delivered artifact associated with the investigated Windows endpoint. The available static evidence was not sufficient to conclusively classify the artifact as malware. A definitive determination would require additional controlled dynamic analysis and deeper reverse engineering.

Skills Demonstrated

Network traffic analysis · Packet analysis · DNS/TLS/HTTP investigation · Windows/Active Directory traffic analysis · Kerberos and SAMR analysis · IOC identification · HTTP object extraction · Artifact handling · Initial malware analysis · Evidence-based reporting

Project Outcome

The project demonstrates a structured security-investigation workflow: establish the affected endpoint, correlate protocol-level evidence, identify suspicious communications, preserve and extract an artifact, perform safe initial analysis, and report conclusions with appropriate uncertainty.